Inference for Two Way Tables

IS381 - Statistics and Probability with R

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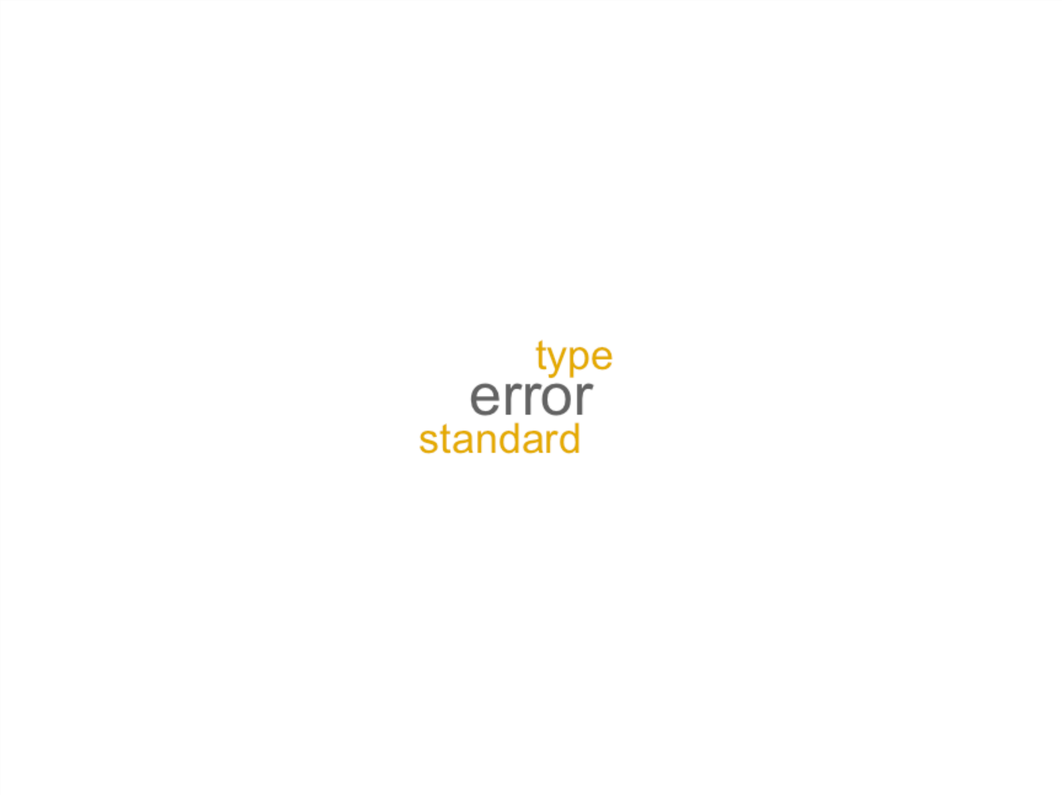
One Minute Paper Results

What was the most important thing you learned during this class?



What important question remains unanswered for you?

Weldon's dice

- Walter Frank Raphael Weldon (1860 - 1906), was an English evolutionary biologist and a founder of biometry. He was the joint founding editor of Biometrika, with Francis Galton and Karl Pearson.

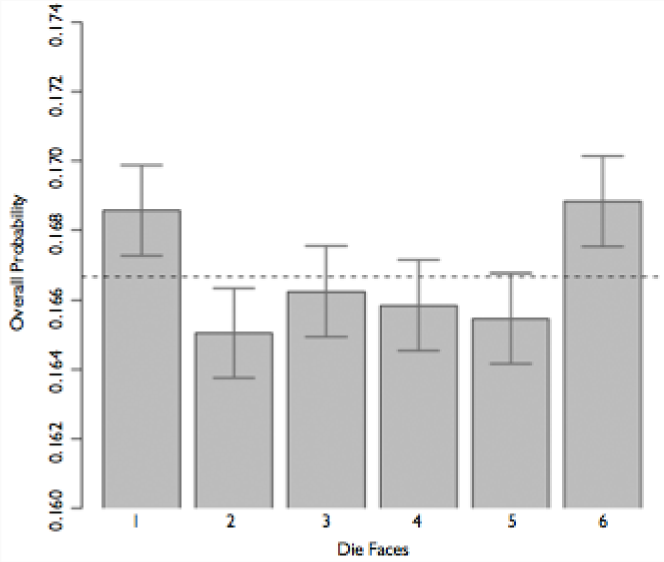
- In 1894, he rolled 12 dice 26,306 times, and recorded the number of 5s or 6s (which he considered to be a success).

    - It was observed that 5s or 6s occurred more often than expected, and Pearson hypothesized that this was probably due to the construction of the dice. Most inexpensive dice have hollowed-out pips, and since opposite sides add to 7, the face with 6 pips is lighter than its opposing face, which has only 1 pip.

Labby's dice

In 2009, Zacariah Labby (U of Chicago), repeated Weldon’s experiment using a homemade dice-throwing, pip counting machine. http://www.youtube.com/watch?v=95EErdouO2w

- The rolling-imaging process took about 20 seconds per roll.
    - Each day there were ~150 images to process manually.
    - At this rate Weldon’s experiment was repeated in a little more than six full days.



Summarizing Labby's results

The table below shows the observed and expected counts from Labby's experiment.

Outcome	Observed	Expected
1	53,222	52,612
2	52,118	52,612
3	52,465	52,612
4	52,338	52,612
5	52,244	52,612
6	53,285	52,612
Total	315,672	315,672

Setting the hypotheses

Do these data provide convincing evidence of an inconsistency between the observed and expected counts?

- H_0: There is no inconsistency between the observed and the expected counts. The observed counts follow the same distribution as the expected counts.
- H_A: There is an inconsistency between the observed and the expected counts. The observed counts **do not** follow the same distribution as the expected counts. There is a bias in which side comes up on the roll of a die.

Evaluating the hypotheses

- To evaluate these hypotheses, we quantify how different the observed counts are from the expected counts.

- Large deviations from what would be expected based on sampling variation (chance) alone provide strong evidence for the alternative hypothesis.

- This is called a *goodness of fit* test since we're evaluating how well the observed data fit the expected distribution.

Anatomy of a test statistic

- The general form of a test statistic is:

$$\frac{\text{point estimate} - \text{null value}}{\text{SE of point estimate}}$$

- This construction is based on

  1. identifying the difference between a point estimate and an expected value if the null hypothesis was true, and
  2. standardizing that difference using the standard error of the point estimate.

- These two ideas will help in the construction of an appropriate test statistic for count data.

Chi-Squared

When dealing with counts and investigating how far the observed counts are from the expected counts, we use a new test statistic called the chi-square (χ^2) statistic.

$$\chi^2 = \sum_{i=1}^{k} \frac{(O - E)^2}{E}$$

where k = total number of cells

Outcome	**Observed**	**Expected**	$\frac{(O-E)^2}{E}$
1	53,222	52,612	$\frac{(53{,}222-52{,}612)^2}{52{,}612} = 7.07$
2	52,118	52,612	$\frac{(52{,}118-52{,}612)^2}{52{,}612} = 4.64$
3	52,465	52,612	$\frac{(52{,}465-52{,}612)^2}{52{,}612} = 0.41$
4	52,338	52,612	$\frac{(52{,}338-52{,}612)^2}{52{,}612} = 1.43$
5	52,244	52,612	$\frac{(52{,}244-52{,}612)^2}{52{,}612} = 2.57$
6	53,285	52,612	$\frac{(53{,}285-52{,}612)^2}{52{,}612} = 8.61$
Total	315,672	315,672	24.73

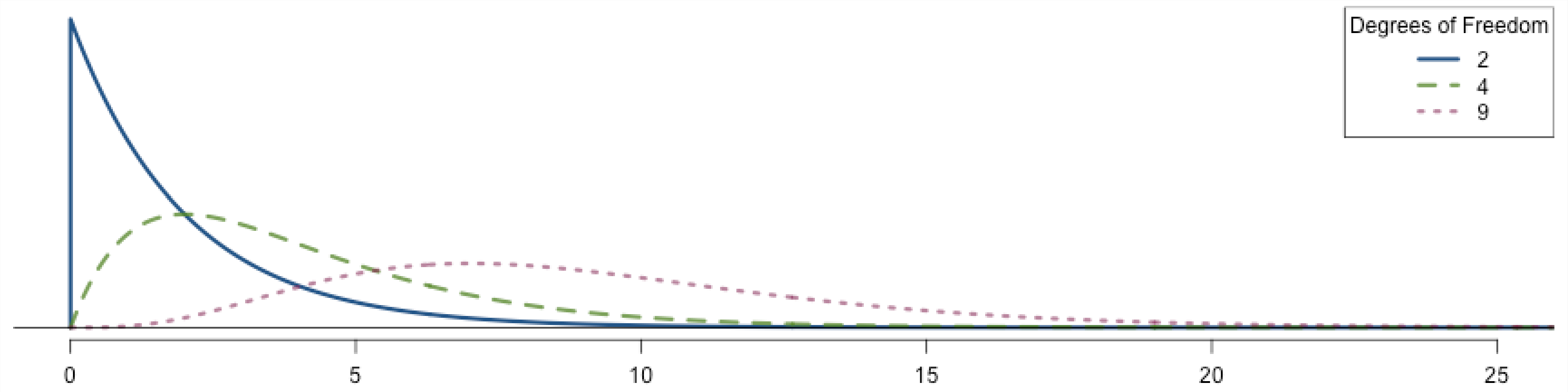
Chi-Squared Distribution

Squaring the difference between the observed and the expected outcome does two things:

- Any standardized difference that is squared will now be positive.
- Differences that already looked unusual will become much larger after being squared.

In order to determine if the χ^2 statistic we calculated is considered unusually high or not we need to first describe its distribution.

- The chi-square distribution has just one parameter called **degrees of freedom (df)**, which influences the shape, center, and spread of the distribution.

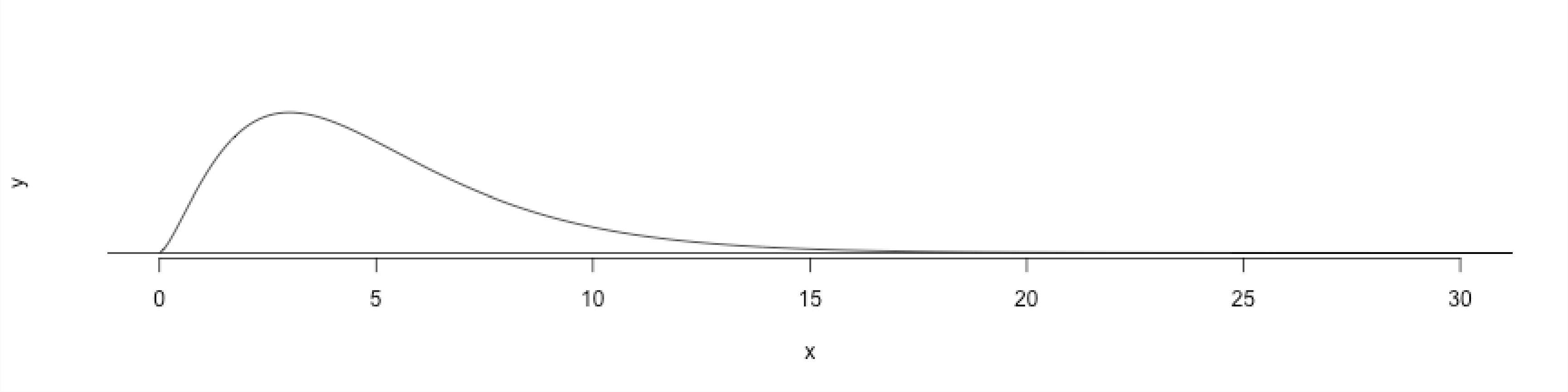


Degrees of freedom for a goodness of fit test

When conducting a goodness of fit test to evaluate how well the observed data follow an expected distribution, the degrees of freedom are calculated as the number of cells (k) minus 1.

$$df = k - 1$$

For dice outcomes, $k = 6$, therefore $df = 6 - 1 = 5$



p-value = $P(\chi^2_{df=5} > 24.67)$ is less than 0.001

Turns out...

- The 1-6 axis is consistently shorter than the other two (2-5 and 3-4), thereby supporting the hypothesis that the faces with one and six pips are larger than the other faces.

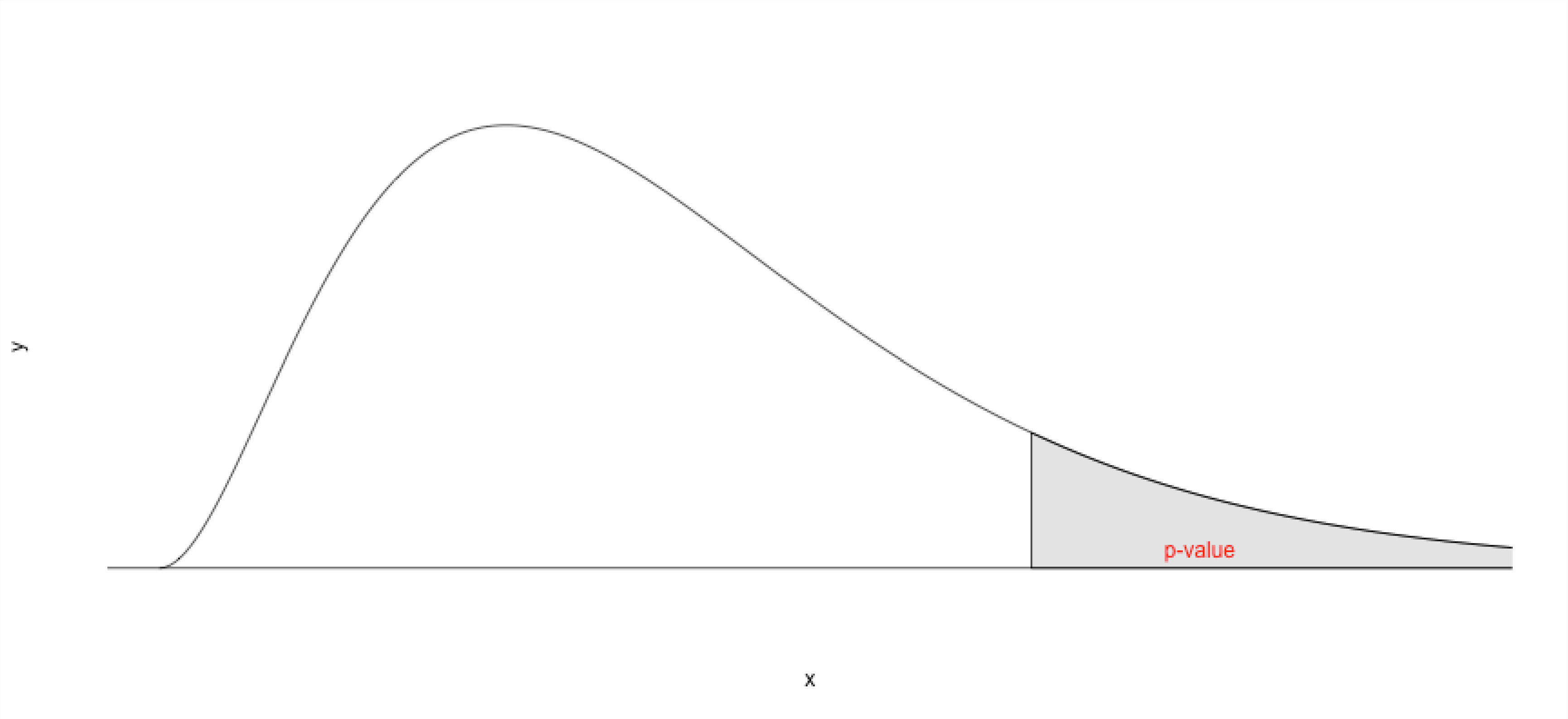
- Pearson's claim that 5s and 6s appear more often due to the carved-out pips is not supported by these data.

- Dice used in casinos have flush faces, where the pips are filled in with a plastic of the same density as the surrounding material and are precisely balanced.

Recap: p-value for a chi-square test

- The p-value for a chi-square test is defined as the tail area **above** the calculated test statistic.

- This is because the test statistic is always positive, and a higher test statistic means a stronger deviation from the null hypothesis.



Independence Between Groups

Assume we have a population of 100,000 where groups A and B are independent with $p_A = .55$ and $p_B = .6$ and $n_A = 99,000$ (99% of the population) and $n_B = 1,000$ (1% of the population). We can sample from the population (that includes groups A and B) and from group B of sample sizes of 1,000 and 100, respectively. We can also calculate $\hat{p}$ for group A independent of B.

```
propA <- .55    # Proportion for group A
propB <- .6     # Proportion for group B
pop.n <- 100000 # Population size
sampleA.n <- 1000
sampleB.n <- 100
```

```
pop <- data.frame(
    group = c(rep('A', pop.n * 0.99),
              rep('B', pop.n * 0.01) ),
    response = c(
        sample(c(1,0),
               size = pop.n * 0.99,
               prob = c(propA, 1 - propA),
               replace = TRUE),
        sample(c(1,0),
               size = pop.n * 0.01,
               prob = c(propB, 1 - propB),
               replace = TRUE) )
)
sampA <- pop[sample(nrow(pop),
                    size = sampleA.n),]
sampB <- pop[sample(which(pop$group == 'B'),
                    size = sampleB.n),]
```

Independence Between Groups (cont.)

$\hat{p}$ for the population sample

```r
mean(sampA$response)
```

```
## [1] 0.549
```

$\hat{p}$ for the population sample, excluding group B

```r
mean(sampA[sampA$group == 'A',]$response)
```

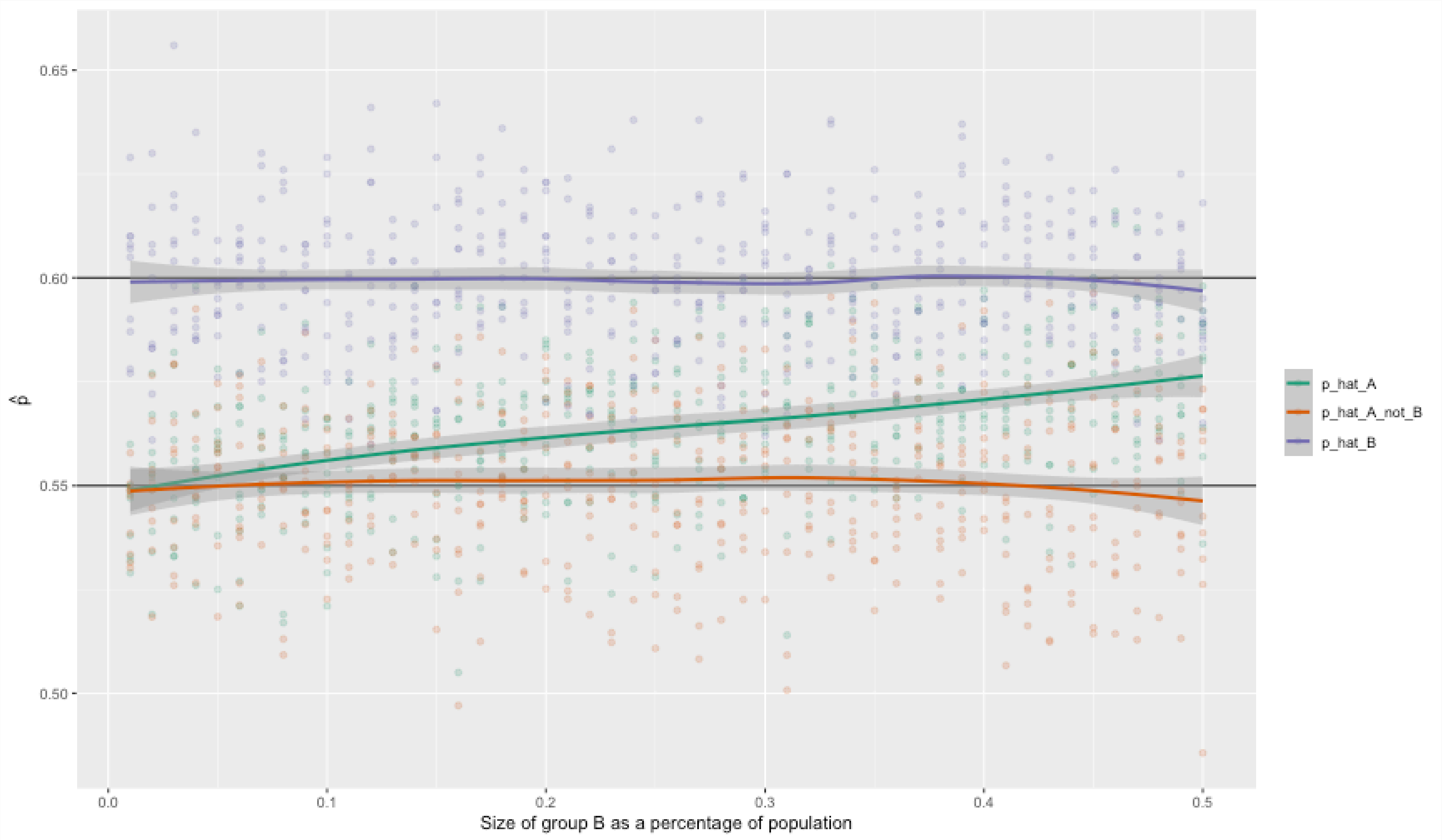
```
## [1] 0.5498489
```

$\hat{p}$ for group B sample

```r
mean(sampB$response)
```

```
## [1] 0.6
```

Independence Between Groups (cont.)



One Minute Paper

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2. What important question remains unanswered for you?

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